Background Suppression Network With Attention Collapse Inhibited Transformer for Optical Remote Sensing Object Detection

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***Abstract*— Object detection for remote sensing imagery (RSI) has been extensively exploited in practical applications. However, similar and multiscale objects in RSI, especially small objects, pose challenges to RSI object detection methods. Particularly, existing approaches ignore irrelevant background in RSI leading to hardship in discriminative feature extraction, resulting in instances of false positive (FP) and false negative (FN) of similar objects. In this article, we propose an irrelevant background suppression network (IBS-Net), which employs the structure of a convolutional neural network (CNN) in series with a Transformer to efficiently capture local and global information in images to respond the challenge of multiscale object detection. Primarily, a background detach module (BDM) is designed behind the backbone to suppress the irrelevant background and enhance the foreground to minimize the interference of irrelevant background for object detection. Furthermore, a composite-sampler (C-S) is devised to sample the vectors describing the foreground and the context, which expands the limited receptive field of the detector to better distinguish similar objects. Especially, considering that transformer-based object detection methods suffer from an attention collapse issue that leads to a degradation of the network representation. An attention collapse inhibited transformer (ACI-former) is presented by designing a partial residual connection, which induces the network to perceive more target information and reduces the loss of small target features thus improving the detection accuracy of small objects. Ultimately, we have conducted related experiments on two benchmarks, which demonstrate that our method has achieved prominent results compared with other mainstream detection methods.**

***Index Terms*— Object detection, one-stage, optical remote sensing, transformer.**

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I. Introduction

OBJECT detection consisting of two subtasks, object localization and object classification, is of paramount importance in achieving image recognition and target extraction, which has a broad development prospect in remote sensing fields, such as natural disaster monitoring [1], maritime investigation [2], military strikes [3], and urban construction [4]. In contrast to natural images, remote sensing imagery (RSI) are taken by sensors on spacecraft from a bird's-eye perspective, which have the characteristics of a complex background, a relatively small proportion of target pixels, and multiscale targets, bringing great challenges to image parsing, especially object detection.

The emergence of deep learning supplies a new horizon for object detection. Deep learning-based models are trained with a large number of data from different scenarios and comprehend target features adaptively, therefore have gained prominence in accuracy and robustness. Commonly, deep learning-based methods are categorized into two-stage and one-stage detection based on different processing modes. Classical two-stage approaches, such as Faster Region Proposal Network (Faster RCNN) [5] demonstrate excellent detection performance in natural images, which is also widely employed in the field of remote sensing. However, due to the traits of multiscale targets in RSIs, the first stage of Faster RCNN only extracts proposals of a fixed size which impacts the detection performance. In response, Zhu et al. [6] proposed a dynamic feature fusion network to enrich spatial information of multiscale targets for remote sensing, which assists in improving the robustness of the detector to multiscale targets. In addition, Wei et al. [7] proposed a feature enhancement feature pyramid network (FE-FPN) to adapt to the issue of scale diversity in RSIs further to avoid the disadvantage of information loss.

Compared with the two-stage object detection, the YOLO [8] approach, which pioneered one-stage object detection, divides the images into grid cells, then determines whether the centers of the targets fall into the cells, and directly predicts the targets' categories and bounding boxes. Respectively, YOLOv7 [9], a variant of YOLO, adopts a fine-grained label assignment strategy and is well adapted to the imbalance of positive and negative samples caused by a relatively small proportion of target pixels in RSIs. However, YOLOv7 is

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not sufficiently accurate in detecting objects in RSIs due to the lack of an advanced multiscale strategy in the detection process. Subsequently, Zhang [10] proposed Drone-YOLO for RSIs, which employs a three-layer path aggregation feature pyramid network (PAFPN) structure and a sandwich fusion mechanism to fuse high-level low-resolution features with low-level high-resolution features. Furthermore, Li et al. [11] presented a bidirectional feature pyramid network (FPN) to combine features across scales for images captured by unmanned aerial vehicles (UAVs) and introduced an additional detection head for small objects.

However, the above approaches, whether single-stage or two-stage detection approaches are limited by the size of the convolutional kernel in convolutional neural network (CNN) methods. The CNN-based methods process decent local feature extraction capability but cannot capture long-distance dependency between pixels, leading to inadequate global information extraction. This limitation becomes more pronounced while considering objects that occupy a large area of the RSI, since large-scale object detection relies on the global receptive field [12] to provide complete target information, including the target's shape information and location information. Then, the Transformer-based [13] approaches are known gradually for their superior ability in capturing global information [14], and self-attention, as its essential mechanism, is employed to calculate the correlation between pixels in the image to capture the long-distance dependence among pixels. Nevertheless, many background pixels that exist in RSIs are not related to the targets, so the self-attention mechanism evaluating the correlation between each pair of pixels leads to a large degree of computational redundancy. Accordingly, Chen et al. put forward a bitemporal image transformer (BIT) [15], which represents the imagery as a few tokens to reduce the feature mapping of the redundant parts and model the relationship among the major pixels. In addition, Lai et al. [16] designed the target awareness enhancement (TAE) module to generate target prototypes to assist the network in generating high-quality tokens, dismissing the transformer's computation for low-quality tokens.

Nevertheless, although the previously stated Transformer-based methods alleviate the computation to some extent with superior, they still pay insufficient attention to exact contextual information, i.e., the correlation between the targets and the background of the targets' adjacent domain, which plays an important role in solving the issues of false positive (FP) and false negative (FN) of similar objects due to the interference of irrelevant background during the detection process. For instance, the bridge (BG) (target) in Fig. 1 lacks discriminative features with the highway (background), which makes it difficult for the detector to distinguish between the two, thus resulting in instances of FPs and FNs. Some recent work has also investigated the above issue. For example, Biswas and Tešić [17] utilize support-guided debiased contrastive learning to reduce the influence of FN examples. Reconstructed feature alignment network (RFA-Net) [18] designs a sparse feature reconstruction module to achieve instance feature alignment. Biswas and Tešić [19] apply an asymptotic domain adaptive scheme for local and global feature alignment. The strategy

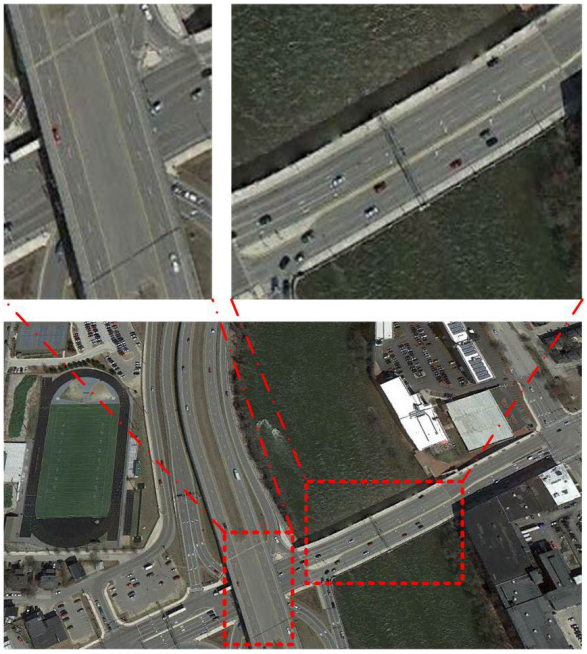


Fig. 1. Data comes from the NWPU dataset, and as shown, the characteristics of BGs and highways are similar and difficult to distinguish.

we employ is to further enhance the discriminative ability of the detector in cooperation with appropriate contextual information. For example, the stream surrounding the BG in Fig. 1 guides the detector to distinguish it from the highway.

Moreover, recent studies have revealed that the Transformer-based approaches have the shortcoming of attention collapse [20], i.e., as the structure of the Transformer deepens, weight matrices of the network tend to be the same, which prevents the performance of the Transformer model from being improved. Therefore, the performance of object detection, especially of small objects, is further restricted.

In order to address the above-mentioned drawbacks of poor detection performance of similar and small objects, we propose a competitive irrelevant background suppression network (IBS-Net) in this article, which adopts a serial structure of CNN and Transformer to improve the detection performance of multiscale objects in RSIs. After the backbone, we design a background detach module (BDM) to evaluate the quantity of information of feature vectors and utilize the property of the informative formula to enlarge the distinctiveness between targets and irrelevant backgrounds, which suppresses irrelevant redundant backgrounds while enhancing the target features. Irrelevant backgrounds are redundant backgrounds at a distance from the target locations due to the unbalanced distribution of foreground and background in remote sensing images. Subsequently, the composite-sampler (C-S), which extracts vectors describing foreground and context evaluated by informative scores, utilizes the context information to remedy the issue of unsatisfactory accuracy of similar objects and small object detection in RSIs. Ultimately, this article presents the attention collapse inhibited transformer (ACI-former) structure to mitigate the disadvantage of attention collapse to further enhance the detection performance. The ACI-former breaks the symmetry of the network by designing partial residual connections in the late stage of the Transformer

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blocks cascading process to update the deep network, which enhances the representation of the network and boosts the recall rate of the network thereby reducing the loss of small object information. The main contributions of this article are as follows.

1) In this article, we propose a novel IBS-Net consisting of CNN and Transformer in tandem, which addresses the drawbacks of limited receptive fields and feature loss in object detection methods and reduces the false alarms of FP and FN. Relevant experimental results show that our method outperforms other state-of-the-art object detection methods, especially for small and similar objects.
2) A BDM inserted behind the backbone has been devised to restrain irrelevant background. Subsequently, a C-S is designed to extract vectors depicting foreground and essential context evaluated by informative scores, which separates instances and expands the receptive fields of the detector to improve the detection accuracy of similar objects.
3) Especially, an ACI-former is presented to strengthen the representation ability of the network to further settle the issue of attention degradation in the late stage of the Transformer blocks cascading process, which motivates the network to be aware of more targets and reinforces recall for object detection. Therefore, the ACI-former increases object detection accuracy, especially for small objects.

II. Related Works

A. Exploration of Contextual Information

The context information refers to the connection between the target pixels and pixels of the surrounding background. The context information supports the network to improve its understanding of the targets by providing additional information. In the task of object detection, the background around the targets provides some valuable hints for inferring the classes of the targets.

Radar cross section (RX) [21] as an anomalous object detection method utilizes covariance to describe the correlation between the target and the surrounding elements, and a threshold is set to identify anomalous targets. Li et al. [22] proposed attention-to-context CNN (AC-CNN) to assist the detector in acquiring both global and local context features. Moreover, an attention-based module and a local context extraction structure are designed to identify a specific kind of target. Xie et al. [23] explored the transformer to utilize the global receptive field to obtain abundant contextual information. Li et al. [24] proposed an attention mechanism module (AMM) to generate joint space-channel features to fully understand the contextual information of remote sensing data. Contextual Transformer (CoT) [25] combined self-attention mechanisms and convolutional operations in the Transformer to capture global and local contextual information in images to improve detection performance. Shi et al. [26] argued that existing CNN-based detectors have only a finite receptive field, making it difficult to apply the global contextual information that is critical for the detection of RSI targets. Therefore, GCFM was designed to adaptively integrate global contextual information using a spatial attention mechanism.

The methods mentioned before utilize the transformer to exploit global contextual information to expand the receptive field of the detector. However, a great deal of redundant information exists in the RSIs. Not all background information is necessary for improving object detection performance, and adding irrational background even impairs detection performance. Therefore, in this article, our approach suppresses the unrelated background and extracts significant context vectors directly.

B. Small Object Detection in RSIs

The detection performance of small objects in RSIs is not ideal in comparison with large objects due to its limited discriminating features and the paucity of samples containing small targets within the dataset. Small objects are targets with a pixel area of less than 32×32 pixels as defined in COCO [27]. Cheng et al. [28] provided an in-depth analysis of small object detection methods for RSIs in the survey. Due to the characteristics of convolutional operation, small objects occupy fewer features and contain a large number of features from the unrelated background at each pixel in the feature map of the deep network, which affects the model detection performance. On the other hand, an imbalance of the number of images existing in remote sensing datasets. For example, small instances in the DOTA dataset make up only 20% of all instances, which causes the detector to focus on large objects.

To address the indigent performance of small object detection, RRNet [29] adopts the method of data augmentation by horizontal flipping and random cropping labeled instances to increase the number of small instances and improve the detection performance of small targets. SuperYOLO [30] fuses multimodal data to enhance small object features. In addition to the way of data augmentation, the super-resolution approaches provide another way for small target detection. Xiaolin et al. [31] enhanced small object features by designing a super-resolution enhancement module. Deconv-RCNN [32], through transposed convolution of low-resolution feature maps, recovers the small target information lack caused by the previous layer, further improving the detection accuracy. hierarchical small object detection network (HSOD-Net) [33] obtains potential target locations through key point prediction, and then purposeful super-resolves the areas corresponding to these targets' locations, thereby avoiding invalid operations in no targeted areas. Furthermore, the approaches based on multiscale feature fusion also enhance the performance of small object detection. Based on the single shot detector (SSD) [34] network, fusion and scaling-based single shot detector (FSSSD) [35] retains its multiscale detection advantages and fuses the features from different layers of the backbone to obtain feature representations with both high resolution and ample semantic information, which greatly amends the small object detection capability of SSD. Biswas and Tešić [36] exploited spatial pyramid pooling and partial cross-stage network to fuse multiscale features. Context-aware detection network (CADNet) [37] introduces a spatial and scale-aware attention module, which directs the network to focus on more valuable

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regions and more appropriate feature scales. Scale selection pyramid network (SSPNet) [38] improves the representation of small targets in the feature pyramid network (FPN) through the scale enhancement module and the scale selection module.

The abovementioned approaches enhance the detection performance of small targets to a certain extent but ignore the advantage of combining the Transformer with CNN. In this article, a serial structure of CNN and Transformer is utilized to capture image local features and global features to improve the detection performance of remote sensing multiscale objects. What is more, the introduction of an ACI-former structure reinforces the recall rate of the network, thereby enhancing the representation of small targets and improving small object detection performance.

C. DETR Series End-to-End Object Detection Algorithms

With the development of object detection technology, many anchor-free methods have been invented, which eliminate the drawbacks of manually designing anchors and the imbalance of positive and negative samples. Detection transformer (DETR) [39], as one of the anchor-free methods, treats object detection as a set prediction and introduces a transformer to object detection first. It achieves performance comparable with Faster RCNN but possesses the drawback of slow convergence and poor performance in detecting small objects. Many scholars have explored DETR and produced many advanced works, Deformable DETR [40], which applies a sparse attention method based on a transformer to reduce computation and speed up the convergence of DETR. Multiscale features in Deformable DETR are also utilized to improve performance, but the computational cost remains a bottleneck. Selective query recollection (SQR-DETR) [41] explores the role of the decoder in transformer structure. Specifically, the decoder of DETR translates object queries into object categories and bounding boxes by applying the multilayer perceptron (MLP) process. Each layer of the decoder accepts the queries of the previous layer and refines it. However, SQR-DETR points out that cascading errors exist in DETR and some valuable queries are discarded. Therefore, SQR-DETR filters the queries of each layer to retain valuable queries. In a recent study, Sparse DETR [42] updating only the tokens that the decoder expects to reference was proposed as a solution, which assists the model in detecting targets efficiently. In addition, applying auxiliary detection loss to selected encoder tokens improves performance while minimizing computational overhead. Based on preliminary experiments, it was found that even if only part of the encoder tokens were updated, the detection performance barely deteriorated.

The previously stated efforts to improve transformer efficiency are mainly focused on accelerating attention operations and improving the robustness of the decoder, with little consideration given to spatial redundancy. The main goal of the transformer-based set prediction (TSP) [43] is to reduce the training cost of DETR. After a lot of analysis, the single-stage approach transformer-based set prediction-fully convolutional one-stage (TSP-FCOS) and the two-stage approach TSP-RCNN are proposed. TSP-FCOS utilizes a binary classifier feature of interest (FOI) [44] module to select positive tokens in images. TSP-RCNN extracts positive proposals in the first stage. Poll and pool (PnP)-DETR [45] proposed the method of reducing spatial redundancy through a new PnP sampling module, and applies this module to build an end-to-end PnP-DETR architecture that adaptively distributes computation spatially to improve computational efficiency.

Our approach solves the spatial redundancy issue of RSIs by extracting foreground and context vectors to detach extraneous background. Also, our approach removes the decoder structure, thereby solving the drawback of slow convergence of DETR training.

III. Methods

In this section, we commence by detailing the proposed network IBS-Net, which consists of a CNN in series with a Transformer, and analyze its network architecture and components in depth. In order to suppress the influence of extraneous background on object detection and to align the features learned by the network with the targets, we elaborate on the design process of the BDM. In this structure, an informative formula was utilized to enlarge the distinctiveness between targets and backgrounds. Subsequently, we delve into the role of contextual information for target detection and design a C-S structure sampling target features with contextual features to identify targets accurately. Moreover, we explore the matter of attention collapse that exists in the Transformer and introduce residual connections in the late stage of the Transformer blocks cascading process, which is capable of updating the deep network and improving the network recall.

A. Proposed Framework

The complete network diagram proposed in this article is exhibited in Fig. 2. The network is composed of CNN and Transformer in series. First, RSIs are fed into the backbone to extract features adaptively. Considering the characteristics of multiscale objects in RSIs, we adopted the FPN structure to preserve small target features as much as possible. After the FPN, we designed the BDM, which aligns the target with the features. The channels of the feature map were compressed, and the foreground features were more prominent after BDM processing while extraneous background was suppressed. Next, a C-S structure is employed to select vectors representing foreground and context, and the foreground and context vectors were concatenated for postprocessing, which can separate instances while reducing redundant calculations for irrelevant backgrounds. What is more, this article presented the ACI-former that alleviates the issue of attention collapse. It improves the network's perception of more target information. Unlike DETR, there is no cross-attention structure in our network, which has proven to be the main reason for slow DETR convergence. As shown in (2), the ACI-former of this article introduced residual connections in the last two blocks of the transformer cascading process to update the deep network. Finally, the prediction head of IBS-Net carried out the set prediction strategy. Which is different from current mainstream object detection heads. The Hungarian loss formula is as follows:

$$\mathcal{L}_{\text{Hungarian}}\left(\bar{y}, \hat{y}\right) = \sum_{i=1}^{N}\left[\mathcal{L}_{\text{class}}^{i,\hat{\sigma}(i)} + 1_{\{c_i \neq \emptyset\}}\mathcal{L}_{\text{box}}^{i,\hat{\sigma}(i)}\right] \quad (1)$$

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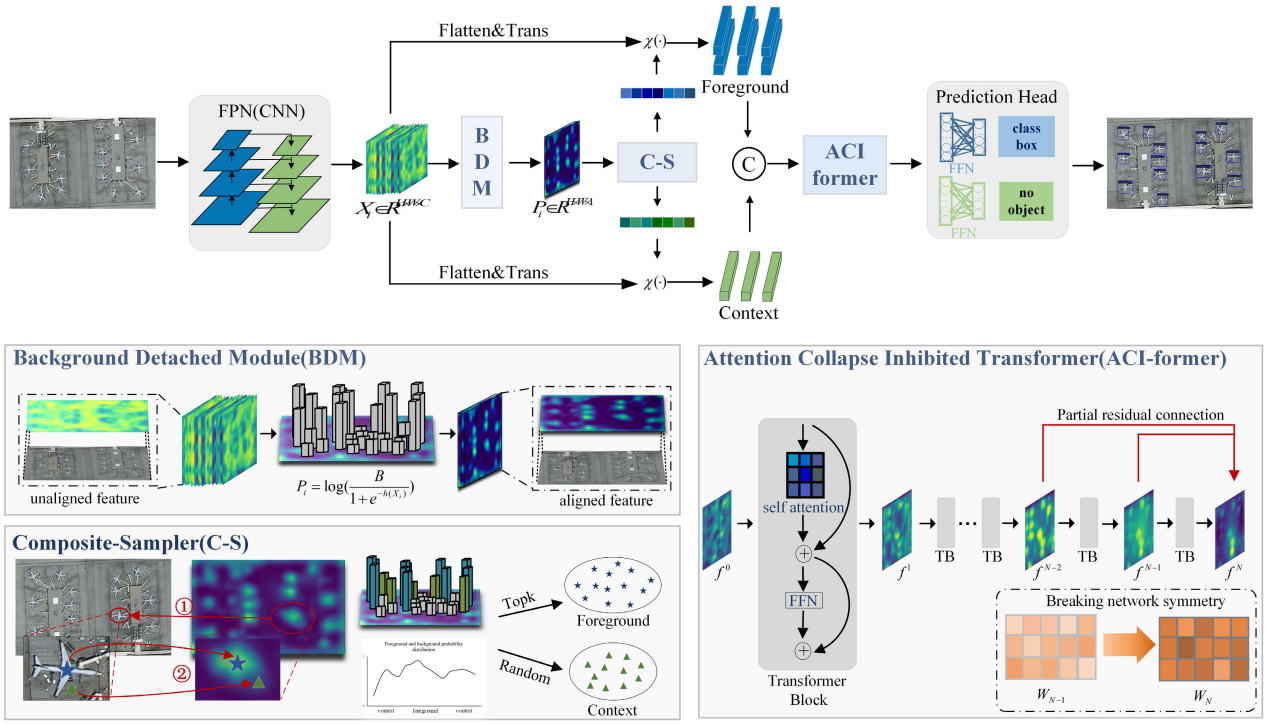


Fig. 2. Pipeline of IBS-Net, RSIs first are fed into the FPN to extract features, then BDM is utilized to suppress irrelevant background after that foreground vectors and context vectors are extracted by harnessing C-S, and finally, ACI-former is presented for global modeling, and then, the prediction head is employed for classification and regression. W_N and W_{N-1} represent the weight matrix of self-attention in the transformer structure. N means the nth transformer block.

where $\bar{y}$ represents the set of ground truth of targets, and $\hat{y}$ denotes the set of predictions of the network. The N in the formula represents the number of prediction boxes, which is usually assigned a large value. N tends to be greater than the true number of ground-truth boxes. Therefore, the number of ground-truth boxes is supplemented to N with no object boxes. And then, the prediction set and the ground truth set can be matched one-to-one. The smallest Hungarian loss of the ground-truth boxes and the prediction box is considered the best match. The formula $\mathcal{L}_{\text{class}}^{i,\hat{\sigma}(i)}$ is expressed as follows:

$$\mathcal{L}_{\text{class}}^{i,\hat{\sigma}(i)} = -\log \hat{p}_{\hat{\sigma}(i)}(c_i) \tag{2}$$

where $\sigma(i)$ is the matching pair selected in the topmost formula

$$\mathcal{L}_{\text{box}}^{i,\hat{\sigma}(i)} = \lambda_{\text{iou}}\, \mathcal{L}_{\text{iou}}\left(b_{\sigma(i)}, \hat{b}_i\right) + \lambda_{L1}\|b_{\sigma(i)} - \hat{b}_i\|_1. \tag{3}$$

In (3), a linear combination of l_1 loss and generalized intersection over union (IoU) loss is combined.

B. Background Detached Module

Experiments have revealed that the feature maps generated from FPN make it difficult to distinguish the difference between the foreground and the background, which exhibits that the FPN framework does not pay enough attention to the target area. It is detrimental to adapt object detection tasks. We designed a BDM module and took the 3-D feature maps as input. It first utilizes convolutional operation to push the detector to increase its focus on the target region and compress the feature map channels and then exploits the informativeness formula we proposed to score the informativeness of the probability distribution of the feature map to realize the alignment of the features. With BDM processing, irrelevant background features are suppressed and target features are enhanced

$$\hat{x}_i = h_{\lambda_i}(x_i) \tag{4}$$

where $h(x)$ stands for convolutional operation. $x_i \in R^{H\times W\times C}$ and $\hat{x}_i \in \mathrm{R}^{H\times W\times 1}$. Different convolutional kernels are represented as λ_i. In our method, we apply a 3×3 convolutional operation. The probability distribution of the feature is represented as $p(x)$

$$p(x) = \frac{e^{\hat{x}_i}}{1 + e^{\hat{x}_i}}. \tag{5}$$

After that, we utilize the information formula to score the features

$$\xi(x) = \log(p(x)) + C \tag{6}$$

where $\xi(x)$ represents the informative score for the feature. C represents the fixed bias, the purpose of which is to guarantee a positive value for the quantity of information.

The applicable information of the feature map is significantly correlated with the score. In the object detection process, the target information are the most important. Features that are not beneficial for the object detection task, such as irrelevant backgrounds, receive low scores due to their insignificant information. Therefore, by using this module, the foreground can be made more prominent and the irrelevant background suppressed.

The informative formula $\xi(x)$ which represents the informative operation on the probability distribution, takes advantage

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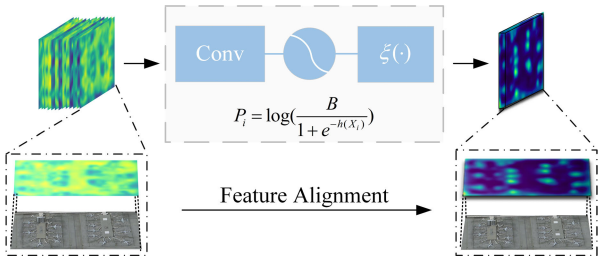


Fig. 3. BDM illustration, BDM includes convolution, sigmoid activation function, and information calculation, which can suppress irrelevant background and align features.

of the fact that the logarithmic function holds a large gradient when the probability distribution is between 0 and 1. Thus, it is possible to enlarge the distinctiveness between foreground and background. The mathematical model mapped by our designed BDM is shown in the following equation:

$$\varphi(x) = \log\left(\frac{1}{1+e^{-h_{\lambda_i}(x_i)}}\right) + C. \tag{7}$$

The module makes it possible to distinguish more clearly between the foreground and unimportant background. As shown in Fig. 3, after BDM processing, the background that is not essential to the feature map's characterization is reduced, allowing the foreground region to be better aligned with the target in the RSI.

C. Composite-Sampler

We introduce a C-S that samples the most important vectors representing targets in the object detection task on the one hand, and the vectors describing context around the target on the other hand to assist the detector in classifying the targets more accurately.

We now dive into analyzing the core components of the C-S structure. After BDM, features are scored by the quantity of information, and the degree of importance of features is ranked as follows:

$$\varphi(F_f) > \varphi(F_c) > \varphi(F_u) \tag{8}$$

where F_f represents the vectors describing foreground, F_c represents the context vector, and F_u represents the irrelevant background vector. $\varphi(\cdot)$ represents the informative equation. After BDM processing, the vectors describing foreground and background are well-separated. The foreground is represented by the image's highlights, as shown in Fig. 4. Our specific approach is illustrated in the C-S module in Fig. 2. The blue pentagram represents the targets of RSIs, and the green triangle represents the context around the target. As can also be seen from the feature map in Fig. 2, the information importance is ranked as $\varphi(F_f) > \varphi(F_c) > \varphi(F_u)$. The feature vectors with the highest information score, i.e., the foreground vectors, are extracted using the top-k sampler. Also, the vectors describing context are extracted using the random sampler. The reason we adopt the top-k sampler is that the target information is crucial in the object detection task and occupies a relatively small percentage of the image. The reason for incorporating a random sampler is to obtain a sparse representation of

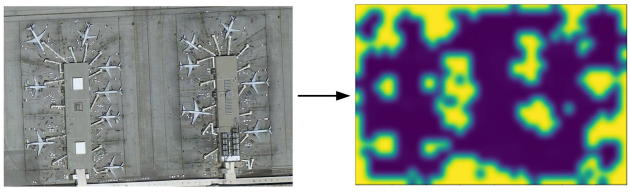


Fig. 4. Binary image representation of foreground–background. The highlighted portion, i.e., the top 200 informative pixels, represents the foreground of the image.

the contextual information, which expands the receptive field without increasing the computational effort

$$\mathbf{F_f} = \psi(\mathbf{g_n})(n = 1, 2, \ldots, N) \tag{9}$$

$$\mathbf{F_c} = \chi(\mathbf{g_m})(m = p, p+1, \ldots, q). \tag{10}$$

$\psi(\cdot)$ in (9) represents the top-k operation. N represents the total number of target vectors. In the design of the top-k sampler, we first rank the informativeness of each pixel in the feature map, record the index of the top N-ranked pixel, and sample the vectors at their corresponding indexes. $\zeta(\cdot)$ in (10) represents a random selection operation. p and q represent the upper and lower bounds of the randomly selected vector range. Specifically, in the design of the random sampler, we set a range (p, q), record the indexes of the pixels whose rank lies within that range, and sample the vector at its corresponding indexes

$$\chi(\mathbf{g_m})(m = p, p+1, \ldots, q) = \text{random}(p, q) \tag{11}$$

$$\psi(\mathbf{g_n})(n = 1, 2, \ldots, N) = \text{top} - k(N). \tag{12}$$

Equations (11) and (12) are further explanations of the top-k and random operations.

In the C-S process, we take the 3-D feature maps with the BDM-processed feature map as inputs, record the indexes of foreground vectors and context vectors of the BDM-processed feature maps, and sample the corresponding positions of the 3-D features. This module initially separates the top 200 pixels as the foreground. The threshold value of 200 is determined in the following way: we map the feature values to a Gaussian distribution, calculate the mean and standard deviation of this distribution, and utilize the 0.683 rule to determine the threshold value, and 200 is the threshold value that we determine by rounding upward. As shown in Fig. 4, where the top 200 pixels are highlighted, and the foreground can be extracted completely. In addition, we consider the vectors ranked 200–500 as context vectors and randomly sample these 300 vectors to reduce the amount of computation. In our approach, we keep only 100 vectors from the 300 context vectors. Finally, we contacted 100 randomly selected context vectors with 200 foreground vectors

$$\mathbf{F} = \mathbf{F_f} \cup \mathbf{F_c} \tag{13}$$

$$\mathbf{F_f} = \left\{\mathbf{f_i} \in \mathbf{R^C}, i = 1, 2, \ldots, N\right\} \tag{14}$$

$$\mathbf{F_c} = \left\{\mathbf{f_j} \in \mathbf{R^C}, j = p, p+1, \ldots, q\right\} \tag{15}$$

where F_f represents the foreground vectors and F_c represents the context vectors. F represents the combination of foreground vectors and context vectors.

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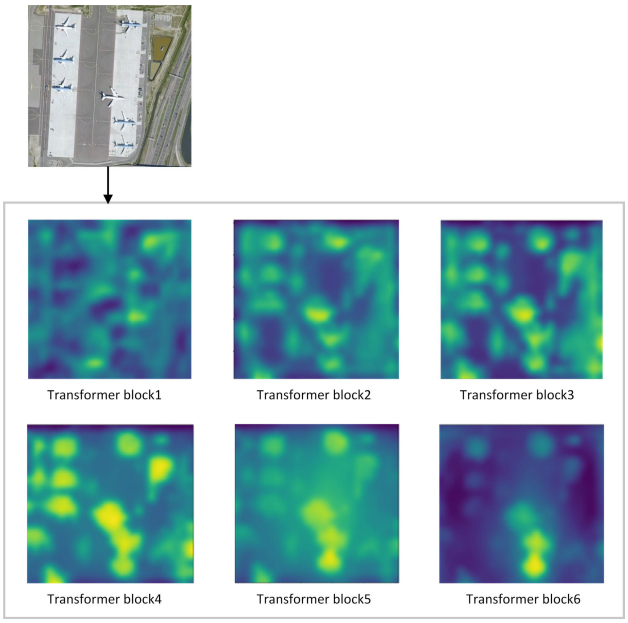


Fig. 5. Visualization of feature maps in Transformer blocks. Starting from block 5, the features begin to degrade.

The C-S module separates the target instances on one hand and removes a lot of unnecessary background from the RSIs on the other hand. It achieves this by extracting foreground and randomly sampled context vectors. In addition to separating the samples, the module also extracts important contextual information from the RSIs, which improves the target detection accuracy and alleviates the computation of the subsequent Transformer structure to some extent.

D. ACI-Former Structure

In this article, we introduce the ACI-former structure for global modeling. Compared with CNNs, the Transformer approach has a larger receptive field and stronger context-shaping capability, which allows for more robust context modeling between any pixel pair in an image. Therefore, we utilize the Transformer structure to achieve object detection more accurately. However, during the block cascading process, we found the disadvantage of feature degradation. This is shown in Fig. 5.

The whole degradation process can be clearly illustrated in Fig. 5. As the blocks cascade, the feature map progresses from block1 to block 4. However, in block 5 and block 6, the feature map deteriorates gradually, which implies that during the later stages of the transformer block cascading process, the network fails to evolve and instead suffers from a decline in the representing capacity of its network

$$\text{Sim} = \frac{\sum_{i=0}^{N}\left(\alpha_i^M - \alpha_i^{M+1}\right)^2}{\sum_{i=0}^{N}\left(\alpha_i^M\right)^2} \tag{16}$$

where α_i^M refers to the ith element of the self-attention weight matrix in the Mth transformer block, and α_i^{M+1} refers to the ith element of the self-attention weight matrix in the $(M+1)$th transformer block. N represents the number of elements in the weight matrix. Equation (16) is utilized to assess the similarity of these matrices between adjacent blocks. Essentially, the larger the difference between the matrices, the higher the value of (16).

TABLE I
SIMILARITY OF THE TRANSFORMER'S ADJACENT INTERLAYER WEIGHT MATRICES. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

block1&2	block2&3	block3&4	block4&5	block5&6
1.85	1.95	**2.34**	1.79	1.71

Starting with block 4, the weight matrix Table I exhibits a progressive rise in similarity. This suggests that only a small number of hidden cells in the deep network change their activation values in response to inputs. Most hidden cells respond consistently to a given input. The results are consistent with the findings in Fig. 5, which emphasizes the issue of attention collapse that exists in the Transformer. The ACI-former structure, which is suggested in this study, introduces residual connections when the Transformer block weight matrix starts to degenerate in order to overcome the issue of attention collapse. In the processing of ACI-former, We take the 2-D vector of $[L, C]$ as input, where L denotes the length of the feature vector and C denotes the number of channels of the feature vector. The residual connections choose the ideal route to disrupt network symmetry adaptively. Compared with the topology with no residual connections, partial residual connections break the network symmetry and increase its representing ability. Therefore, the ACI-former structure reduces the loss of significant features, improves the recall of the network, and assists in improving the accuracy of object detection, especially for small objects.

IV. EXPERIMENTS

A. Evaluation Metrics

If a classifier predicts a sample as positive and it is positive, it is called a true positive (TP). On the other hand, if the classifier predicts a sample as positive but it is actually negative, it is called a FP. When the classifier correctly predicts that a sample is negative, it is called a true negative (TN). However, if the classifier predicts a sample as negative but it is actually positive, it is called an FN.

In object detection tasks, boxes that overlap with a certain amount of similarity, known as the IoU score, are classified as TP instances or FP instances. FN instances occur when the model fails to detect a ground-truth bounding box. The AP value changes with different IoU thresholds, with a higher IoU threshold resulting in a lower AP value. Higher AP indicates better performance at a given IoU threshold. The object detection evaluation indexes include precision, recall, and mean average precision (mAP). The accuracy calculation formula is as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{17}$$

The recall calculation formula is as follows:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{18}$$

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Fig. 6. Example of detection results for the NWPU dataset, including categories: AP, SP, ST, BD, TC, BC, HB, BG, and VH.

mAP is a crucial performance indicator for object detection tasks. It combines the AP values of various classes and averages the AP of all categories. To calculate mAP, the IOU threshold typically is set to 0.5 or 0.95, and then calculate the Aps at different IoU thresholds for each category, averaging these AP values.

B. Dataset Description

We compared our proposed network to classical and advanced remote sensing object detection methods on the NWPU and DIOR datasets, achieving state-of-the-art results.

1) The NWPU VHR-10 dataset is a collection of geospatial objects explicitly used for remote sensing research purposes, which contains ten categories of objects, namely, AP, Ship (SP), storage tank (ST), baseball diamond (BD), tennis court (TC), basketball court (BC), golf field (GF), harbor (HB), BG, and vehicle (VH). The dataset consists of 800 RSIs with ultrahigh resolution, taken from Google Earth and Vaihingen datasets. Experts have manually annotated all the images in the dataset. In our experiments, we used 650 positive sample images, out of which 80% were used for training, and 20% were used for testing.
2) The DIOR Dataset is a comprehensive remote sensing object detection dataset, comprising 190 288 instances, spanning 20 object classes. The object classes include airplane (AL), airport (AT), baseball field (BF), BC, BG, chimney (CM), dam (DM), expressway-service-area (EA), expressway-toll-station (ES), ground track field (GC), GF, HB, overpass (OP), SP, stadium (SD), ST, TC, train station (TS), VH, and windmill (WM). The experiment in this article follows the default division method of the DIOR dataset. For our experiment, 5862 images are used as the training set while the remaining 5863 images are used as the test set.

TABLE II
RESULTS OF ABLATION EXPERIMENTS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

baseline	baseline*	forground	context	ACI-former	mAP(%)
✔					92.51
	✔				90.76
	✔	✔			94.86
	✔	✔	✔		95.82
	✔			✔	91.98
	✔	✔	✔	✔	**96.21**

C. Experimental Settings

Our experiments were conducted under the Ubuntu 9.4.0 system and in the hardware platform of NVIDIA RTX2080Ti with 11-GB memory. The experiment environment is Pytorch 1.6. In experiments where experiments are located in Pytorch 1.6, we set the batch size to 2 per GPU, the initial learning rate to 0.01, and the HYBRID optimizer was utilized to optimize the model parameters.

D. Experimental Analysis

We conduct experimental validation on the corresponding DIOR and NWPU datasets and our approach produces state-of-the-art results on the DIOR dataset as well as the NWPU dataset. Our ablation experiments were conducted on the NWPU dataset, and we contrast our network with advanced single-stage detector and transformer RSI object detection approaches on the NWPU and DIOR datasets.

1) Ablation Experiment: In order to adjust to the significant difference in target scale in RSIs, we add the FPN structure to our framework. After that, we apply the BDM to suppress extraneous background and obtain the informative fraction of the feature vector.

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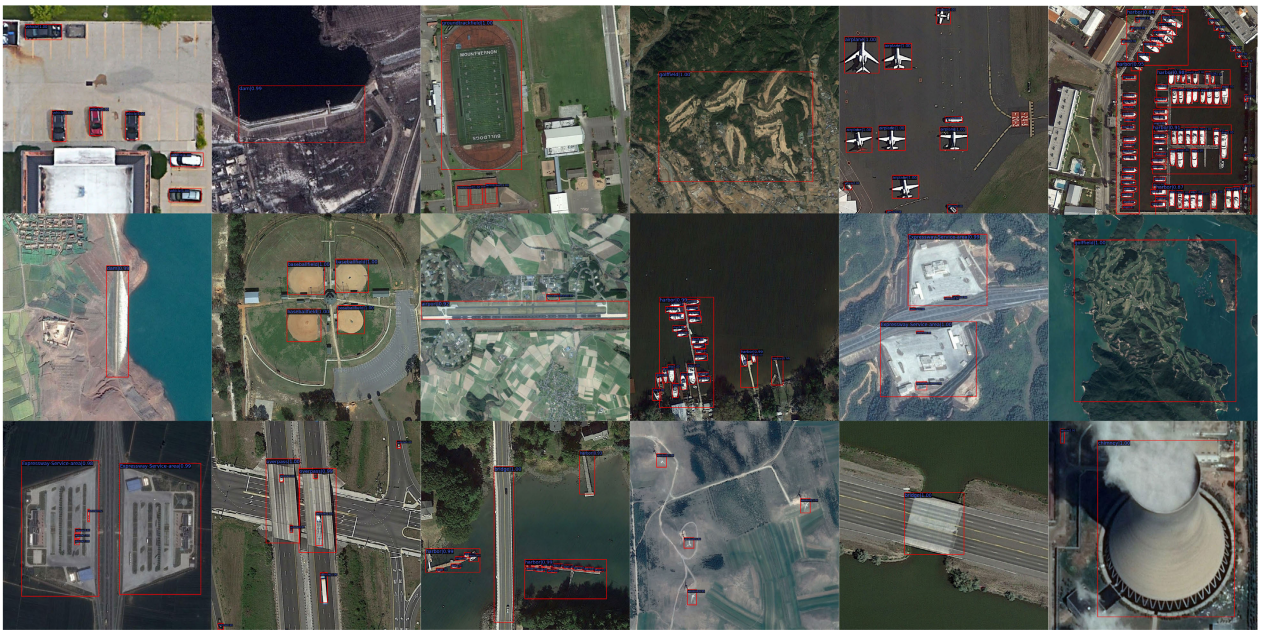


Fig. 7. Example of detection results for the DIOR dataset, including categories: airplane (AL), AT, BF, BC, BG, CM, DM, EA, ground track field (GC), GF, HB, OP, SP, TC, VH, and WM.

TABLE III

EFFECT OF ABLATION EXPERIMENTS ON THE *Bridge* CLASS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

foreground	context	ACI-former	AP(%)
✔			84.37
✔	✔		90.99
✔	✔	✔	**94.64**

TABLE IV

ABLATION EXPERIMENTS ON CONTEXT THRESHOLD DETERMINATION

	mAP	AP(BG)
Foreground(top 200 pixels)	94.858	84.367
Foreground+context(top300)	95.473	86.717
Foerground+context(top200+rand((200,500),100))	95.822	90.987
Foreground+context(top200+rand((200,600),100))	95.263	89.965
Foreground+context(top500)	96.023	90.202

TABLE V

ABLATION EXPERIMENTAL RESULTS OF THE NUMBER OF TRANSFORMER LAYERS

one layer	three layer	four layer	six layer
93.47	94.42	94.44	94.86

We choose the top 200 ranked vectors evaluated by the BDM as foreground vectors in the C-S module to extract the foreground as fully as possible. The 100 context vectors were randomly selected to add to the necessary discriminative information surrounding the target, and the ACI-former structure was applied after combining the foreground and context vectors.

It is worth noting that we add the FPN structure behind the backbone of the network in order to retain the small target features. In Table II, baseline* represents the network structure with the decoder structure removed, on which we carry out all subsequent ablation experiments. Table II contains the findings of the ablation experiment.

The baseline holds a mAP of 92.5% across ten categories of targets in the NWPU dataset, as shown in Table II. Following the BDM's assessment, the C-S module was solely utilized to accurately extract the foreground (94.86%). The mAP is 95.82% after adding the context vector, which is 1.05% greater than only extracting the foreground. By adding only the ACI-former structure to the baseline*, the mAP can reach 91.98%, which is 1.2% higher than the baseline*. We replace the transformer with an ACI-former on the basis of the network after extracting the foreground and context vectors to achieve the best mAP of 96.21%. Our top mAP exceeds the baseline by 3.7%.

An experimental investigation on the AP of the BG category in the NWPU dataset was conducted to confirm the efficacy of our module. The BG category is more challenging to identify in RSI object detection than other categories. One of the key factors is that not many distinguishing traits exist in the category of BGs, and the presence of numerous comparable targets in RSIs, such as OPs, highways, and other similar structures hampers the recognition of BGs. For auxiliary detection, our method augments the detection process using context information. With the help of our technology, the detector can more easily perceive and recognize different item types during the detection process by adding crucial context information.

As shown in Table III. The AP of the BG is only 84.37% when the foreground vectors are only extracted for detection. The AP is 90.99% after the context vectors are included. The AP of the BG is increased by 6.62% as compared to

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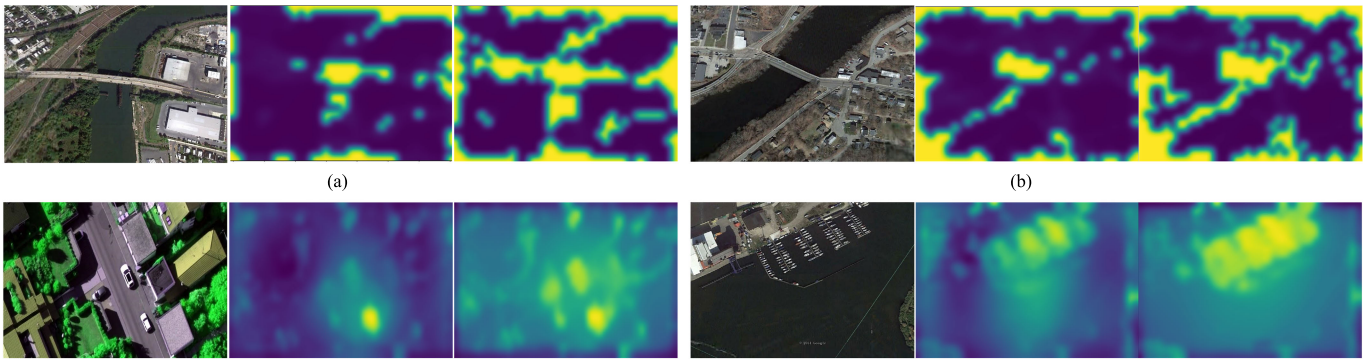


Fig. 8. Visual analysis of network features. The two sets of figures in the first row (a) and (b) represent the foreground schematic and foreground + context schematic, respectively. The second line (c) and (d) represents the difference between the feature maps output by the ACI-former structure and the traditional Transformer structure.

TABLE VI

COMPARISON OF OUR ALGORITHM WITH OTHER ADVANCED RSI OBJECT DETECTION ALGORITHMS ON NWPU DATASET. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED

Method	AP	BD	GF	**VH**	SP	TC	HB	**ST**	BC	BG	mAP	epochs	FPS	GFlops
CenterNet[46]	98.9	96.9	93.2	89.0	74.0	74.5	71.4	87.0	95.6	48.1	82.9	45	29	87.305
Cascade R-CNN [47]	**99.0**	97.4	**100.0**	93.0	91.4	91.0	95.3	87.5	93.1	85.9	93.4	20	12	125.304
EfficentNet [48]	98.7	97.0	99.1	93.9	86.3	92.7	86.2	98.6	91.0	83.3	92.7	70	14	73.877
Faster RCNN	**99.0**	97.0	99.3	92.9	91.3	92.6	92.3	97.3	91.1	86.9	94.0	60	14	97.606
FCOS [49]	98.9	97.1	**100.0**	92.0	88.2	87.0	92.1	94.8	90.8	81.8	92.3	200	16	85.275
YOLOv3	98.5	**97.9**	99.7	93.3	83.6	**99.6**	94.7	99.2	93.0	86.5	94.6	300	38	83.061
DETR	98.8	97.8	99.0	94.0	92.1	91.3	91.3	73.9	93.3	93.3	92.5	300	12	57.679
Swin [50]	98.7	97.8	99.9	90.5	90.3	94.8	91.2	95.0	88.8	89.8	93.7	100	12	94.636
Deformable DETR	98.7	94.3	89.9	**96.6**	92.8	95.7	**100.0**	93.5	86.5	92.6	94.1	60	10	84.602
LSK-Net[51]	**99.0**	97.8	99.7	92.9	**95.4**	94.6	88.6	96.1	96.3	94.4	95.5	50	13	72.391
Ours	98.6	96.3	**100.0**	96.2	91.0	94.7	94.0	**99.9**	**96.4**	**95.6**	**96.3**	20	15	89.417

merely extracting foreground vectors. Finally, the AP of the BG category hit 94.64% after the ACI-former was utilized to replace the transformer structure, which demonstrates the value of incorporating context vectors into our approach for similar object detection and the potency of the ACI-former structure. The ACI-former increases the detector's recall rate to the targets, which significantly impacts AP improvement. In Table IV, threshold 200 is our statistically derived threshold. On this basis, the receptive field is expanded to assess the impact of context boundary settings. When the context boundary is set to 600 pixels, the mAP decreases, suggesting that random sampling reduces the amount of useful context information while introducing irrelevant context. Notably, sampling all the top 500 pixels results in a slight increase in mAP. To balance accuracy with the computational load of the subsequent Transformer layers, random sampling is employed. For the category of BG we are concerned with, the AP is boosted accordingly with the addition of context vectors. We have taken the random sampling to balance the relationship between the accuracy and the amount of the subsequent Transformer computation.

To confirm the efficacy of the ACI-former mechanism, we conduct an ablation experiment on the number of layers of the transformer, as shown in Table V. In the previous analysis, we demonstrated that the transformer structure has the drawback of attention collapse from the fourth layer. The results in the table show that after the third layer, increasing the number of transformer layers improves the object detection accuracy, but the effect is minimal. As shown in Fig. 8, the above row represents two sets of examples (a) and (b) of the impact of contextual information on feature maps. The leftmost represents the original image, the middle represents the feature map with foreground information, and the rightmost represents the feature map with added contextual information.

It could be seen that after adding context, the features are more abundant, so it is easier to identify the target. The bottom row represents two sets of examples Fig. 8(c) and (d) of the feature map variation of the ACI-former structure proposed in this article compared to the classic transformer structure. The leftmost represents the original image, the middle represents the feature map output by the classic transformer, and the rightmost represents the feature map output by the ACI-former structure. It can be seen that our ACI-former structure increases the recall of the network, so target features are more prominent.

2) Comparative Experiments: On the NWPU and DIOR datasets, we compare our approach against the most recent advanced RSI object detection approaches and obtain state-of-the-art results.

It can be seen from the experimental results in Table VI that our method has achieved good detection results for small

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TABLE VII
COMPARISON OF OUR ALGORITHM WITH OTHER ADVANCED RSI OBJECT DETECTION ALGORITHMS ON DIOR DATASET. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED

Method	**AL**	AT	BF	BC	BG	CM	DM	EA	ES	GC	GF	HB	OP	SP	SD	ST	TC	TS	VH	WM	mAP	epochs	FPS
EfficentNet	69.0	4.4	73.4	32.3	5.5	41.9	0.6	5.2	17.6	37.5	1.1	7.2	13.4	49.6	35.2	65.1	68.5	3.8	31.5	1.6	28.5	70	19
YOLOv3	72.2	29.2	74.0	78.6	31.2	69.7	26.9	48.6	54.4	31.1	61.1	44.9	49.7	**87.4**	70.6	68.7	87.3	29.4	48.3	78.7	57.1	300	56
FCOS	61.1	**82.6**	76.6	**87.6**	42.8	80.6	64.1	79.1	67.2	82.0	79.6	46.4	57.8	72.1	64.8	63.4	85.2	62.8	43.8	**87.5**	69.4	200	28
Faster RCNN	91.0	67.2	91.7	81.8	39.6	89.3	50.8	63.5	57.3	73.9	75.0	53.4	62.6	77.6	86.9	76.9	87.1	50.9	68.6	74.2	71.1	60	23
CenterNet	61.2	80.1	73.7	81.4	45.2	75.8	64.8	**81.2**	**76.5**	79.5	**79.7**	47.2	59.3	72.6	70.5	53.7	82.6	55.9	49.1	77.8	68.4	45	31
Cascade R-CNN	91.1	71.1	91.9	82.8	42.7	90.0	57.9	65.2	58.1	77.2	77.3	55.4	63.7	78.6	85.5	81.3	88.0	56.6	71.5	75.0	73.0	20	18
Swin	91.4	68.7	87.5	85.4	43.5	88.2	58.9	64.4	60.1	75.6	78.5	60.1	62.2	73.1	87.2	73.6	89.4	58.3	72.5	73.9	73.5	100	20
DETR	90.2	61.7	90.5	85.2	40.7	78.7	60.2	72.1	57.3	73.4	70.5	68.2	65.3	72.4	89.2	71.5	79.4	64.3	68.5	70.9	73.2	300	21
LSK-Net	91.0	77.8	**92.9**	83.8	**53.4**	**92.2**	65.1	68.7	60.0	**83.8**	74.1	60.0	63.9	78.5	**90.3**	**81.8**	**90.7**	61.3	**73.6**	76.3	76.0	50	22
Deformable DETR	**92.4**	64.6	87.6	82.3	42.5	72.6	63.3	79.1	54.3	77.3	68.5	**69.6**	**69.4**	74.6	88.5	73.5	77.4	**68.2**	71.3	72.3	76.5	60	13
Ours	91.8	68.2	84.6	83.6	40.7	73.6	**68.2**	76.4	58.3	78.2	69.4	67.5	68.4	76.3	86.2	74.6	81.4	64.6	69.4	74.8	**76.7**	20	19

objects and similar objects compared with other advanced methods. For ground track fields, BCs, and BG s, our methods achieve optimal results. For VH and STs, our method yields suboptimal results. And on the NWPU dataset, the mAP of our method reached 96.3%. In Fig. 6, we also visualize the detection results of our algorithm on the NWPU dataset.

As can be seen from Table VII, our method is compared with the current mainstream first-stage algorithm, classic two-stage algorithm, and transformer-related algorithm on the DIOR dataset. Our method achieves state-of-the-art results on the DIOR dataset. The mAP of our method reached 76.7%. In Fig. 7, we also visualise the detection results of our algorithm on the DIOR dataset.

In Tables VI and VII, we have achieved optimal and suboptimal results for the large-scale object categories marked with underlines, the small-scale objects marked with bold lines, and the similar objects marked with dotted lines. Moreover, our method achieves the highest accuracy without consuming much training time and inference time. Since our method which extracts the foreground and context pixels from the original feature map, discarding the redundant background pixels degrades the interference of the redundant background while accelerating model convergence, thus improving the accuracy of object detection and speeding up the training.

V. CONCLUSION

In this article, we propose IBS-Net for the RSI object detection task, which adopts the serial structure of CNN and Transformer to capture local and global information of the image to adapt to the characteristics of remote sensing multiscale objects. Primarily, we insert the BDM behind the backbone to suppress the large amount of irrelevant background in the RSI in order to demote the interference of irrelevant background in RSI object detection. Subsequently, we design the C-S to extract vectors describing foreground and context, which expands the detector's receptive field to distinguish similar objects. In addition, we introduce the ACI-former structure, which alleviates the issue of attention collapse present in the classic Transformer and improves the network's representation capability, especially reducing the loss of small object information. Our approach improves the detection performance of similar objects and multiscale objects, especially small objects. Applying our method to the NWPU and DIOR datasets achieves state-of-the-art performance compared to other advanced methods. However, the inclusion of a transformer block cascade after the CNN backbone leads to higher model complexity, and the application of a module for sorting and sampling after the FPN leads to higher Flops. Therefore, the lightweight design of the network is a particularly important part to consider in our future work.

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